Duolingo Growth & Marketing Operations Analytics

Case Study / Product Operations / User Operations / Growth Operations / General Operations

As Duolingo's audience grew, did engagement and monetisation strengthen too? Across 2023-2025, daily activity and monetisation grew faster than MAU. This case translates that pattern into priorities for product, user and growth operations.

12 quarters | Public data | Tableau | 2023-2025 | March 2026

Open the published interactive Tableau dashboard

1. Business Question, Data & Method

Audience growth alone gives an incomplete view of operating performance. This project compares reach with daily activity, paid subscribers and revenue to identify what an operations team should monitor and investigate next.

Business questions

1. How did audience, daily activity and monetisation change across 12 quarters?

2. Did DAU, paid subscribers and revenue grow faster than MAU?

3. What does the historical competitor context add?

4. Which operational priorities need further internal evidence?

Data & method

12 quarters: Q1 2023-Q4 2025. Official Duolingo IR / SEC disclosures.

Compare DAU/MAU, QoQ, YoY and indexed growth (Q1 2023 = 100).

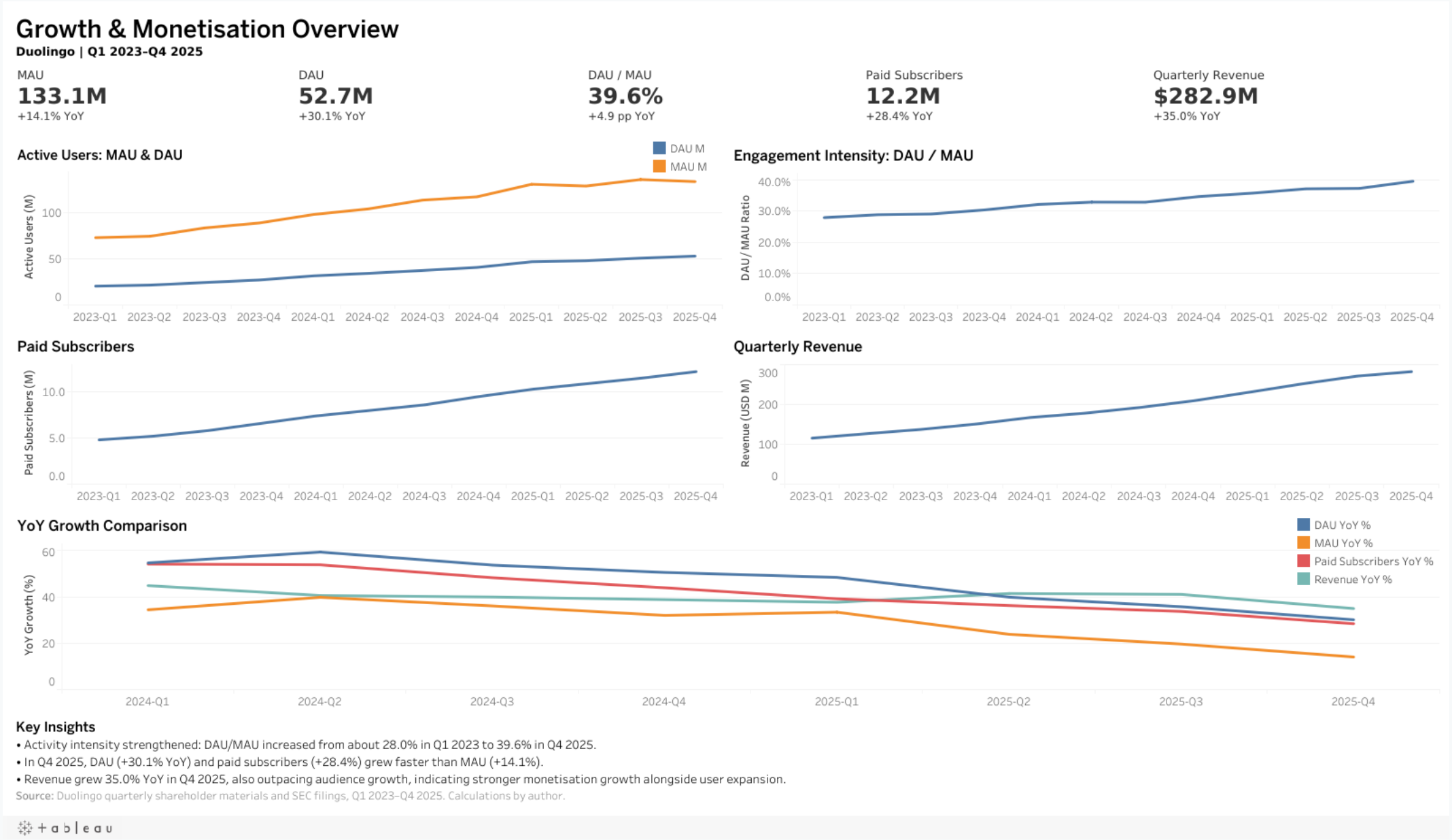
Website: February 2026. Social: March 2026. Historical context only.

Source alignment and SQL / Python checks support two final Tableau pages.

Sources: Duolingo IR / SEC. Definitions and full links: methodology, source manifest and speaker notes

2. Dashboard - Growth & Monetisation Overview

[Final published Tableau view. Click to open the interactive dashboard]



3. Dashboard - Growth Diagnostics & Strategic Context

[Final published Tableau view. Click to open the interactive dashboard]

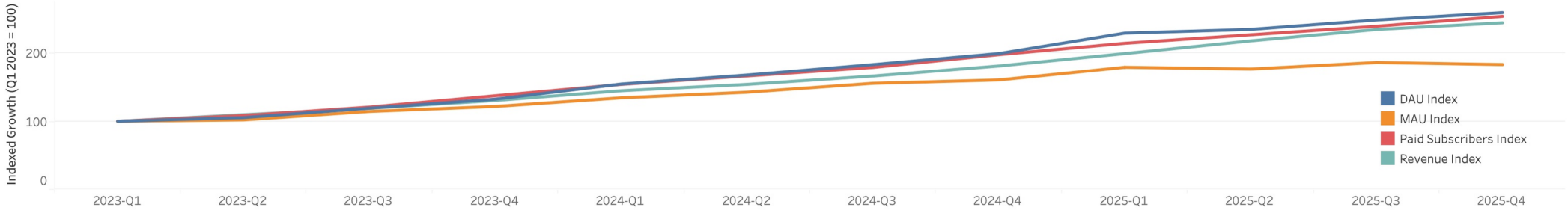
Growth Diagnostics & Strategic Context

Duolingo | Q1 2023–Q4 2025

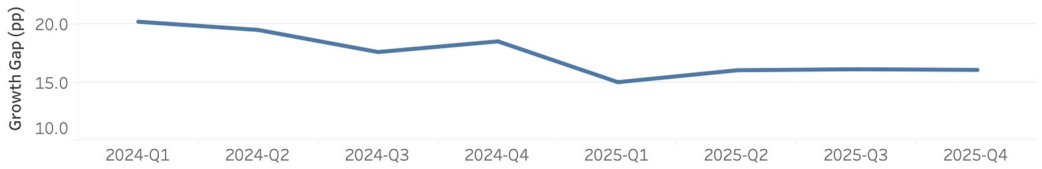
Q4 2025 growth gaps vs MAU

Engagement Growth Gap	Paid Subscriber Growth Gap	Revenue Growth Gap
+16.1 pp	**+14.4 pp**	**+20.9 pp**
DAU YoY − MAU YoY	Paid Subscribers YoY − MAU YoY	Revenue YoY − MAU YoY

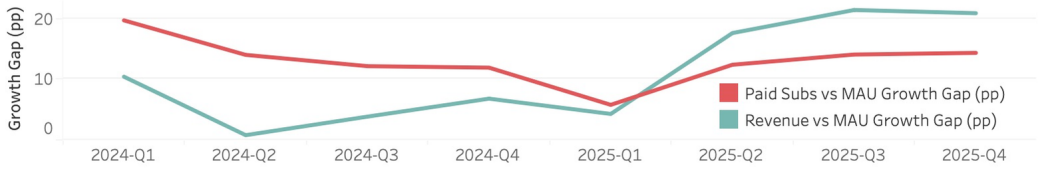
Indexed Growth Comparison



Engagement Growth Gap: DAU vs MAU

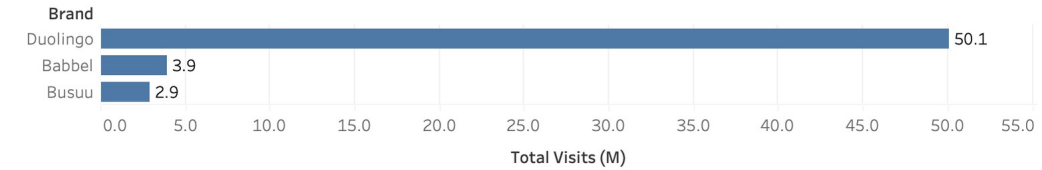


Monetisation Growth Gaps vs MAU



Competitive Digital Context — Feb–Mar 2026

Website Traffic Snapshot — Feb 2026



Social Audience Snapshot — Mar 2026

	Platform		
Brand	Instagram	TikTok	YouTube
Duolingo	4.80M	17.00M	6.61M
Babbel	0.45M	0.56M	0.33M
Busuu	0.41M	0.86M	0.06M

Operational Priorities

- **Protect engagement quality:** investigate the drivers behind DAU growth outpacing MAU and monitor whether stronger activity intensity is sustained.
- **Strengthen premium value communication:** test clearer subscription value messaging and targeted trial opportunities rather than assuming faster revenue growth reflects better conversion.
- **Balance growth with user control:** maintain configurable engagement settings and monitor user response as activity and monetisation continue to scale.

Sources: Duolingo quarterly shareholder materials and SEC filings, Q1 2023–Q4 2025. Competitive snapshots: website traffic, Feb 2026; social audiences, Mar 2026. Competitor snapshots are contextual only; original dated captures are unavailable.

4. Key Findings

Engagement

DAU grew 160% versus MAU 83% from Q1 2023 to Q4 2025. DAU/MAU rose from 28.0% to 39.6% (+11.6 pp), indicating deeper aggregate activity.

Monetisation

Paid subscribers grew 154% and quarterly revenue 145%, both faster than MAU. The data supports further review of monetisation drivers.

Growth mix

By Q4 2025, MAU growth moderated to 14% YoY while DAU, paid subscribers and revenue grew 30%, 28% and 35%.

Competitive context

The inherited February/March 2026 snapshots show greater Duolingo digital reach among the three observed brands. This is context only, with unverified historical captures.

Sources: Duolingo IR / SEC. Definitions and full links: methodology, source manifest and speaker notes

5. Operational Implications

01

Protect engagement quality

Investigate the drivers behind DAU growth outpacing MAU and monitor whether stronger activity intensity is sustained.

02

Strengthen premium value communication

Test clearer subscription value messaging and targeted trial opportunities rather than assuming faster revenue growth reflects better conversion.

03

Balance growth with user control

Maintain configurable engagement settings and monitor user response as activity and monetisation continue to scale.

Sources: Duolingo IR / SEC. Definitions and full links: methodology, source manifest and speaker notes

6. Method & Limitations

How the analysis was built

• Official disclosures cover six corporate metrics across 12 quarters. Sources and definitions are documented.

• Compare period growth, QoQ, YoY, DAU/MAU and indexed growth. Rate gaps use percentage points.

• SQL and Python checks reconcile the 16 derived fields and preserve unavailable comparisons as missing.

• Two completed Tableau dashboards are the final visualisation. Both language versions use the same views.

Analytical guardrails

• Public aggregates cannot measure cohort retention or a free-to-paid funnel.

• Competitor snapshots have no original dated captures and provide context only.

• DAU/MAU describes aggregate activity. Paid and revenue growth do not isolate conversion or pricing.

• Relationships are descriptive. Recommendations have no measured business impact.

The project delivers operational priorities and a monitoring framework. Proposed reviews and experiments require internal data and validation.

Sources: Duolingo IR / SEC. Definitions and full links: methodology, source manifest and speaker notes